



Question pool

Kognitive Systeme (Technische Universität München)



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Cognitive Systems: Exam questions

1 Cognition

1. Describe the Chinese room argument
2. Explain what is the purpose of the Chinese room argument
3. Describe the Turing test
4. Explain what is the robotics paradox
5. Explain the difference between cognitivist cognitive systems and emergent cognitive systems
6. Name seven cognitive capabilities that a system must possess in order to be self-reliant and adaptive, and interact with its environment
7. Name the four different lobes in the cerebrum and explain what are their main cognitive functions
8. Assume that you have a doll company. Your new series features dolls that are more realistic than any toy produced until now. However, the doll is highly unsuccessful and testers describe it as creepy. Name and explain the phenomenon that is taking place
9. A company claims that it can make you smarter by unlocking the 90% unused capacity of your brain. Explain whether this is possible or not
10. Define what is a cognitive system
11. Define cognition
12. Name the components of a cognitive system
13. Name two applications that require cognition
14. Draw the cycle of cognitive processing
15. Name the two schools of thought which deal with the mind-body problem. For each of them, describe their approach to the problem
16. Explain What kinds of tasks robots/computers perform better than humans, and which ones they perform worse. Give an example for each of them.
17. Explain why birds sing using an ultimate explanation and using a proximate explanation

2 NRP and cognitive architectures

1. Name four advantages of executing experiments with virtual robots instead of physical ones.
2. Explain what is the neurorobotics platform and explain the advantages of its usage
3. Name two software tools that are integrated in the neurorobotics platform
4. Explain what is the function of buffers in ACT-R
5. Explain what is a module in ACT-R. Name two features of modules in ACT-R
6. Explain what is a production in ACT-R

3 Human brain

1. Explain whether the following statement is true or false: "Only 10% of the human brain is used"
2. Explain what is the main purpose of the white matter in the cerebrum
3. Name and describe the three main anatomical features of the cerebral cortex
4. Name two distinctive features of human brain that makes it stand out among other species
5. Explain what is the Blue Brain Project
6. Provide the names of the three main components of a biological neuron and describe their main tasks.
7. The cerebral cortex is structured into four different lobes. Name three different lobes present in the cortex. For each of them, name one cognitive function that it implements.
8. Name and describe the two subsystems of the vertebrate nervous system
9. Explain the function of the hippocampus
10. Name two features or properties of the peripheral (autonomic) nervous system
11. Map the nervous system into the perception-cognition-action diagram
12. Explain what is the main purpose of the brainstem. Name the four sections that compose it
13. Describe what is the encephalization quotient
14. Briefly explain the function of the amygdala in the human brain

4 Biological neural networks

1. Name three learning rules based on Hebbian learning
2. Name and explain two paradigms for encoding information in spiking neural networks
3. Name five levels of modeling the dynamics of biological neurons
4. Explain what is the action potential of a biological neuron
5. Describe the Hodgkin-Huxley Model. You may draw a diagram
6. Explain the meaning of the statement "Neurons that fire together, wire together" in the context of learning. Name which type of learning is based on this statement
7. Explain what is neuroplasticity and why it is relevant for the function of the central nervous system
8. Explain what is the refractory period of a neuron. Name the two main phases that it can be split into
9. Explain what is the resting membrane potential of a neuron
10. Name four different activation functions for general analog neuron models
11. Explain what are the main differences between analog neuron models and spiking neuron models
12. Draw a plot of the membrane potential of the neuron over time and label the different sections.
13. Explain the function of the myelin sheath around nerve fibers
14. Solve the differential equation of a LIF neuron for a given scenario.
15. Name two different types of neural plasticity and briefly describe them.

5 Machine learning

1. Name three methods for avoiding the problem of overfitting
2. In the context of machine learning, describe the concept of Max-Pooling.
3. Define "machine learning" with your own words. Provide an example of a robotic task that does **not** require machine learning.
4. Explain what is the advantage of converting the convolutional kernels and inputs of a neural network into the Fourier Space. Explain What is the main challenge for this approach
5. Explain the difference between interpolation and regression.
6. State the problem of using least squares for linear classification. Explain how support vector machines address this problem
7. State the formula of the output of a general analog neuron model and calculate the output given the parameters.
8. Name three learning paradigms. For each of them, describe the type of data needed for training.
9. Explain the method of k-fold cross-validation for partitioning a dataset.
10. Describe the principle of Occam's Razor and state how it is applied in machine learning.
11. Name and describe the role of the two streams of visual processing
12. Name and describe the three conceptual processing layers in the human visual system.
13. Name a problem that can not be solved by a single perceptron with a linear activation function. State briefly or draw the reason for that.
14. State the difference between regression and classification.

6 Reinforcement learning

1. Explain the difference between on-policy and off-policy learning
2. Briefly explain what "bootstrapping value" means in the context of reinforcement learning.
3. Name and describe the two alternating steps that are performed iteratively under the framework of Generalized Policy Iteration.
4. Describe and state the differences between classic conditioning and operant conditioning in the context of associative learning processes
5. State the reason behind using a linear activation function in the output layer of a Deep Q-Network.

Guest lecture

1. Name the two main types of modern neuroimaging. Name one example for each of them
2. Explain the difference between MRI and fMRI, and how they help to better understand neural activity